

Data for Neurocomputing paper, including the
structural and topological analysis of the images
used in this research

César Eduardo Muñoz-Chávez Hermilo Sánchez-Cruz
Elisa Isabel Gómez-Gómez

February 2, 2025

1 Appendix E. Structural and topological analysis of the images.

Tables 1 and 2 outline the comprehensive structural and topological analysis conducted on each of the 66 3D voxelized objects examined in this study. The analysis includes descriptors such as object dimensions, Euler characteristics, and the number of cavities and tunnels. These descriptors are essential for understanding the spatial complexity and topology of the objects, which play a critical role in accurately modeling and analyzing the 3D structures.

Table 1: Objects for testing and its and dimensions (Part 1).

Name	Dimension	Euler	Cavities	Tunnels
Object1	6×5×5	2	0	0
Object2	7×10×9	-4	0	5
Cup	29×25×41	0	0	1
Dragon	42×64×90	0	0	1
Sphere0	65×65×31	1	0	0
Sphere3	65×65×31	3	2	0
Sphere4	65×65×31	3	2	0
Sphere5	65×65×31	1	2	2
Torus	67×65×10	0	0	1
Torush	67×65×10	-1	0	2
Vase	69×53×41	0	0	1
Cheese	102×102×102	-2	1	4
Squirrel	128×128×128	1	0	0
Armchair	128×128×128	-1	0	2
Axe	128×128×128	0	0	1
Batmobile	128×128×128	5	57	130
Bird	128×128×128	-1	0	3
Owl	128×128×128	0	0	1
Dragon	128×128×128	-8	0	9
Elephant	128×128×128	1	0	0
Speakers	128×128×128	-5	0	7
French Horn	128×128×128	-88	0	104
GUN	128×128×128	-7	0	8
Moon	128×128×128	1	0	0
Bear	128×128×128	1	0	0
Sheep	128×128×128	1	0	0
Penguin	128×128×128	1	0	0
Pig	128×128×128	1	0	0
Turtle	128×128×128	1	0	0
Table Lamp	128×128×128	1	0	0
Bolt	256×256×256	1	0	0
Chair	256×256×256	1	0	0
Chameleon	256×256×256	3	0	0
House	256×256×256	-256	0	2258

Table 2: Objects for testing and its and dimensions (Part 2).

Name	Dimension	Euler	Cavities	Tunnels
Ship	$256 \times 256 \times 256$	-8	0	22
Skateboard	$256 \times 256 \times 256$	0	0	1
T34 Soviet Tank	$256 \times 256 \times 256$	-143	0	149
Axolotl	$256 \times 256 \times 256$	-1	0	2
Plane	$256 \times 256 \times 256$	1	0	0
Car	$256 \times 256 \times 256$	1	0	0
Creeper	$256 \times 256 \times 256$	1	0	0
Giraffe	$256 \times 256 \times 256$	1	0	0
Karambit	$256 \times 256 \times 256$	0	0	1
Wolf	$256 \times 256 \times 256$	1	0	0
Nintendo	$256 \times 256 \times 256$	760	0	1
Duck	$256 \times 256 \times 256$	1	0	0
Snake	$256 \times 256 \times 256$	0	0	1
Chandelier	$520 \times 520 \times 520$	-349	0	419
Dinosaur	$520 \times 520 \times 520$	1	0	0
Fish	$520 \times 520 \times 520$	-12	0	14
Kangaroo Chair	$520 \times 520 \times 520$	-2082	0	2512
Lioness	$520 \times 520 \times 520$	16	0	1
Red Hand	$520 \times 520 \times 520$	-23	0	46
Sofa	$520 \times 520 \times 520$	-9	0	10
Teleosaurus	$520 \times 520 \times 520$	0	0	1
Tulip Chair	$520 \times 520 \times 520$	1	0	0
Yak	$520 \times 520 \times 520$	-1	0	2
Batman	$512 \times 512 \times 512$	1	0	0
Knife	$512 \times 512 \times 512$	1	0	0
Staircase	$512 \times 512 \times 512$	1	0	0
Sphere	$512 \times 512 \times 512$	1	0	0
Sword	$512 \times 512 \times 512$	3	0	0
L	$512 \times 512 \times 512$	1	0	0
Brick	$512 \times 512 \times 512$	1	0	0
Peak	$512 \times 512 \times 512$	-3	0	4
Pyramid	$512 \times 512 \times 512$	1	0	0